

Lab 2: Multiple Linear Regression

Background & Study Overview:

In this laboratory exercise, we analyze data from a study conducted by *Baty et al. (2006)*, which explored the biological mechanisms underlying the cardioprotective effects of various beverages. Specifically, we investigate how different beverages alter human blood gene expression over time using linear regression models.

1. Experimental Design & Data Collection

Six healthy human subjects participated in a randomized, controlled, cross-over study. Across four independent test days, each subject consumed 500 mL of one of four target beverages:

- **Grape Juice** (Non-alcoholic polyphenol-rich drink)
- **Red Wine** (Alcoholic polyphenol-rich drink)
- **Diluted Ethanol** (40g ethanol control)
- **Water** (Baseline control)

1.1 Administration Schedule

The experimental assignments across subjects and days were structured as follows:

Participant	Day 1	Day 2	Day 3	Day 4
Individual 1	Grape Juice	Red Wine	Alcohol	Water
Individual 2	Water	Grape Juice	Red Wine	Alcohol
Individual 3	Red Wine	Alcohol	Water	Grape Juice
Individual 4	Water	Alcohol	Red Wine	Grape Juice
Individual 5	Grape Juice	Red Wine	Alcohol	Water
Individual 6	Alcohol	Red Wine	Water	Grape Juice

Table 1: Cross-over experimental design matrix for beverage administration.

1.2 Sampling Procedure

For each study day, blood samples were collected alongside standardized nutritional intake at five specific time points:

- **Baseline (0 h):** Prior to beverage ingestion.
- **Post-Ingestion Timepoints:** 1 hour, 2 hours, 4 hours, and 12 hours.

Due to missed appointments, 12 expected blood samples were missing from the schedule. RNA extracted from the remaining 108 blood samples was hybridized onto Affymetrix microarrays to generate high-throughput gene expression profiles.

2. Dataset Structure & Access

The workspace environment contains the `Alldata` list object, loaded into R via:

```
load(file = "Alldata")
```

The primary expression matrix (`Alldata$originaldata`) comprises 22,283 genes (rows) across 130 feature/metadata columns:

- **Columns 1–2:** Sample/Probe Identification numbers and Gene Identifiers.
- **Columns 3–110:** Normalized expression values (log-transformed counts).
- **Remaining Columns:** Extended gene annotation information.

2.1 Workspace Variable Definitions

- `trt1` – `trt4`: Sample ID vectors corresponding to treatment groups (`trt1`: Alcohol, `trt2`: Grape Juice, `trt3`: Red Wine, `trt4`: Water).
- `time_h0` – `time_h12`: Sample ID vectors for each sampling hour (0h, 1h, 2h, 4h, 12h).
- `ind1` – `ind6`: Sample ID vectors partitioned by individual subject.

3. Lab Assignment

In this lab session, focus specifically on probes corresponding to the gene identifier “**RFC2**” (Replication Factor C Subunit 2). Complete the following tasks:

Q1: Outcome Variable Construction

Load `"Alldata"` into R. Identify all rows in `originaldata` where the gene IDENTIFIER (column 2) matches “**RFC2**”. Extract the gene expression values (columns 3–110) for these probes and construct a single composite outcome variable (Y) by computing the mean expression across all identified RFC2 probes for each sample.

Q2: Predictor Variable Encoding

Construct two predictor variables corresponding to each sample:

- `beverage`: A 4-level categorical factor variable (Alcohol, Grape Juice, Red Wine, Water).

- **hours** : A continuous numerical variable denoting the hours elapsed post-ingestion (0, 1, 2, 4, 12).

Q3: Categorical One-Way ANOVA Model

Fit a simple linear regression model targeting the composite RFC2 outcome variable using only the categorical variable **beverage** as a predictor.

Q4: Contrast Estimation & Hypothesis Testing

Derive the Least Squares Estimate (LSE) for the difference in average RFC2 expression between the **Alcohol** and **Water** groups. Report the estimated difference, standard error, and corresponding p -value. Is the observed effect statistically significant at $\alpha = 0.05$?

Q5: Time-Series Gene Expression Visualization

Let $X = (0, 1, 2, 4, 12)'$ represent the sampling timepoints. Compute Y_k , the average RFC2 gene expression across subjects at each timepoint for beverage $k \in \{1, 2, 3, 4\}$. Construct a clear scatter plot of X versus Y_k , overlaying trajectory lines for all four beverages on a single figure with an appropriate legend.

Q6: Polynomial Response Modeling

Fit a multiple linear regression model incorporating both predictors (**beverage** and **hours**). Include a 3rd-degree (cubic) polynomial transformation for **hours** :

$$Y = \beta_0 + \beta_{\text{bev}} \mathbf{X}_{\text{bev}} + \gamma_1 X_{\text{hours}} + \gamma_2 X_{\text{hours}}^2 + \gamma_3 X_{\text{hours}}^3 + \varepsilon$$

Q7: Prediction & Uncertainty Interval

Using the fitted cubic model from Q6, estimate the predicted average RFC2 gene expression for a subject 6 hours after consuming alcohol. Calculate and interpret the 95% prediction interval for this point estimate.